

CHAPTER III

SIGNALS

A. GENERAL PROVISIONS

- 3.01 GENERAL USE OF SIGNALS.—**The signals prescribed in these rules shall be used for controlling the movement of trains in all cases in which exceptions are not allowed by approved special instructions.
- 3.02 KINDS OF SIGNALS.—**The signals to be used for controlling the movement of trains shall be—
- (a) fixed signals,**
 - (b) hand signals,**
 - (c) detonating signals and**
 - (d) flare signals.**
- 3.03 USE OF NIGHT SIGNALS BY DAY.—**The signals prescribed in these rules for use by night shall also be used by day in tunnels and in thick, foggy or tempestuous weather impairing visibility.
- 3.04 PLACING OF SIGNALS AND SIGNAL ARMS; PAINTING OF SIGNAL ARMS.—**
- (1) Fixed signals shall be clearly visible to the Loco Pilots of trains approaching them and shall be placed immediately to the left of or above the line to which they refer, unless otherwise authorised by special instructions.**
 - (2) In the case of semaphore signals, signal arms shall be placed on left hand side of the post as seen by the Loco Pilot of any approaching train to which they refer.**
 - (3)(a) Except as provided for in clauses (b) and (c), signal arms shall be painted with the same colour as the light exhibited in the ‘on’ position with a white bar on the side facing trains to which they refer and white with a black bar on the other side. Such bars shall be parallel with the end of the arms.**

- (b) In the case of a yellow arm, a black bar shall take the place of the white bar on the side facing trains.
- (c) Calling-on arms shall be painted white with a red bar on the side facing trains to which they refer and white with a black bar on the other side.

B. DESCRIPTION OF FIXED SIGNALS

3.05 USE OF FIXED SIGNALS.—

- (1) Except under approved special instructions, all railways shall be equipped with fixed signals as prescribed in these rules.
- (2) The aspects of a semaphore signal shall be displayed by the position of the arm by day and by a light or lights by night.

Note: - In the illustrations given in this Chapter, which are not drawn to scale, the day aspect of the semaphore signal is shown by the position of the arm and the night aspect is shown by the light or lights to the right of the signal concerned.

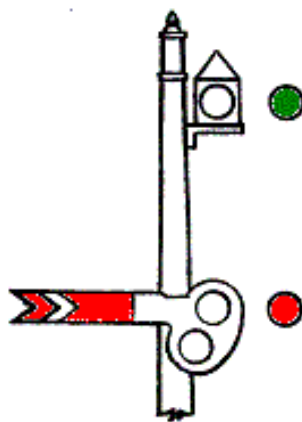
- (3) The aspects of a colour light and position light signal both by day and by night shall be the same and shall be displayed by fixed light or lights.
- (4) The arm of a semaphore signal shall work in –
 - (a) the lower quadrant in two – aspect signalling and
 - (b) the upper quadrant in manually operated multiple-aspect signalling.
- (5) The ‘off’ position of a semaphore signal shall be displayed by day by the inclined position of the arm from 45^0 to 60^0 below the horizontal in case of two-aspect lower quadrant signals and 45^0 or 90^0 above the horizontal in case of multiple-aspect upper quadrant signals.

3.06 DESCRIPTION OF WARNER SIGNALS AND THEIR INDICATIONS.—

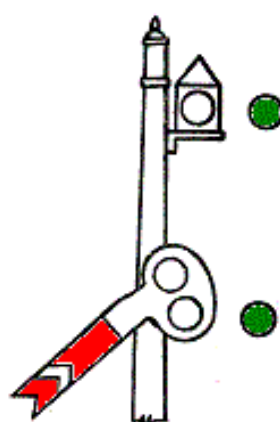
- (1) A semaphore Warner signal has a fish-tailed arm.

- (2) A Warner signal is intended to warn a Loco Pilot—
 - (a) of the condition of the block section ahead, or
 - (b) that he is approaching a Stop signal.
- (3) A Warner signal may be placed either—
 - (a) on a post by itself with a fixed green light 1.5 to 2 metres above it by night, or
 - (b) on the same post below the first Stop signal or the last Stop signal.
- (4) When placed in accordance with clause (b) of sub-rule (3), the variable light of the Stop signal shall take the place of the fixed green light of the Warner signal and the mechanical arrangement shall be such that the Warner signal cannot be taken 'off' while the Stop signal above it is 'on'.
- (5) The aspects and indications of a semaphore Warner signal are shown below: -
 - (a) Semaphore Warner signal in Two-Aspect Signalling Territory – on a post by itself

'On' position



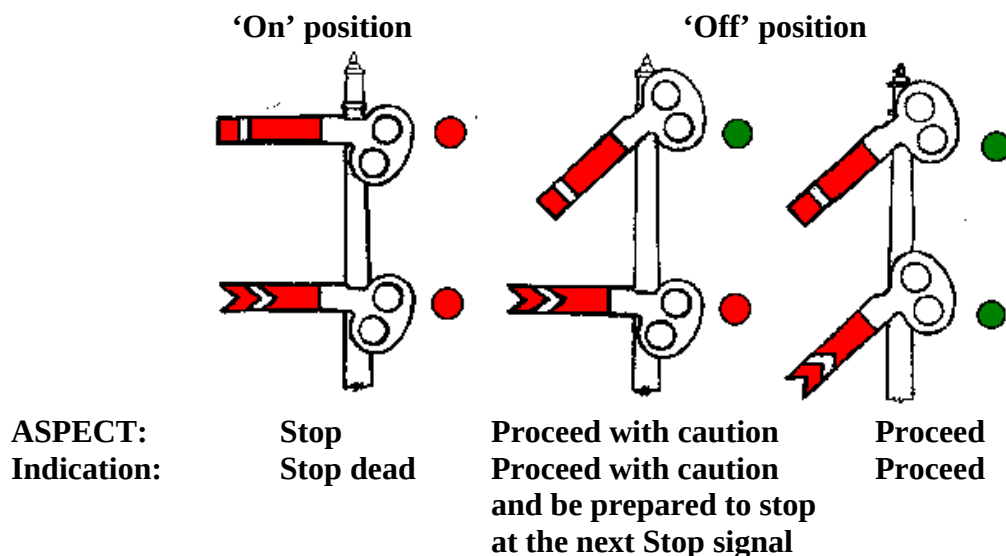
'Off' position



ASPECT: Proceed with caution
Indication: Proceed with caution and be prepared to stop at the next Stop signal

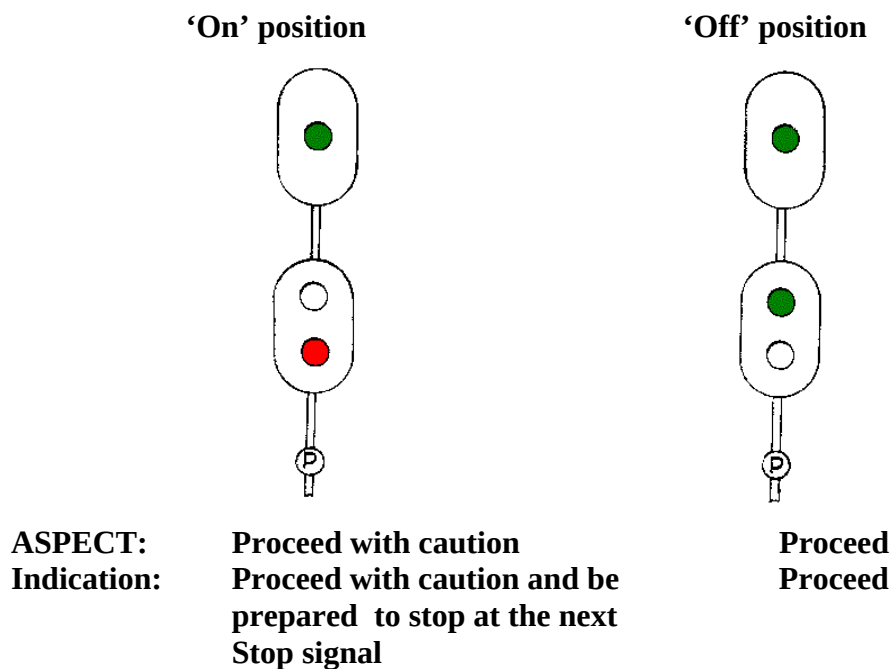
Proceed
Proceed

(b) Semaphore Warner signal in Two-Aspect Signalling Territory-
below a Stop signal

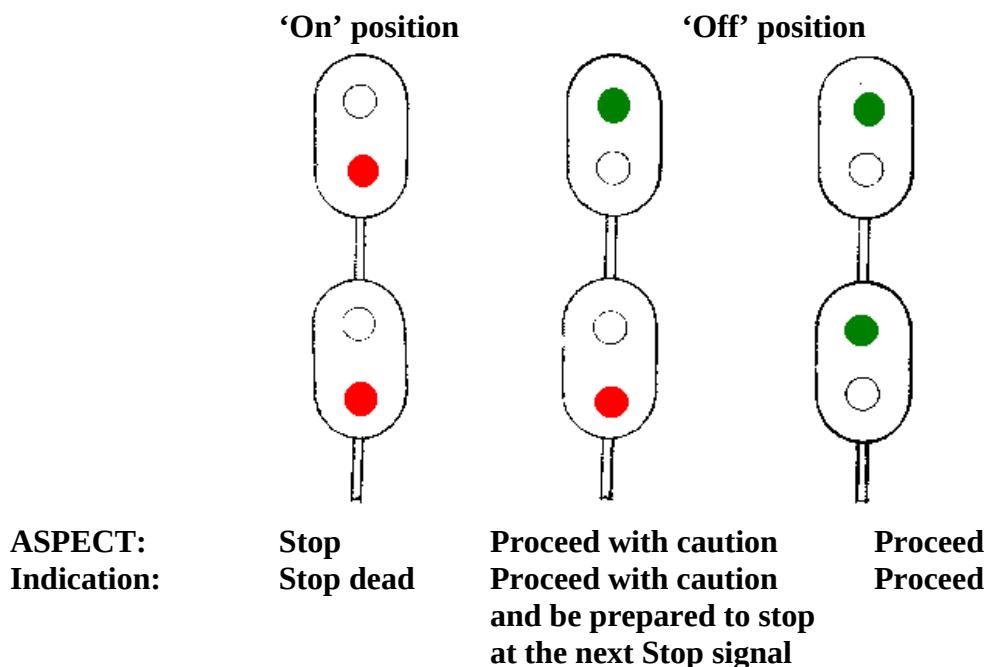


(6) The aspects and indications of a colour light Warner signal are shown below: -

(a) Colour Light Warner signal in Two-Aspect Signalling Territory
on a post by itself



(b) **Colour Light Warner Signal in Two – Aspect Signalling Territory-below a Stop signal**



- (7) A Warner signal with a fixed green light above it by night, on a post by itself, shall be located at an adequate distance in rear of the Stop signal, the aspect of which it pre-warns:

Provided that when such a Warner signal applies to a gate Stop signal it shall not display the ‘Proceed’ aspect unless there is adequate distance between the gate Stop signal and the first Stop signal of the station ahead. The adequate distance in such a case shall never be less than 1200 metres.

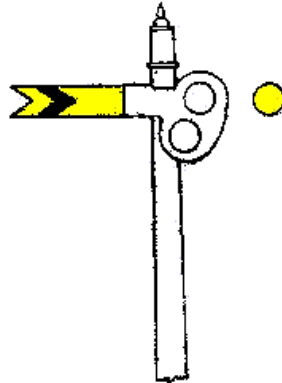
- (8) Where special circumstances justify the use of an unworked Warner, it shall be secured in the ‘on’ position and not be coupled or duplicated for directing purposes.

3.07 DESCRIPTION OF DISTANT SIGNALS AND THEIR INDICATIONS.—

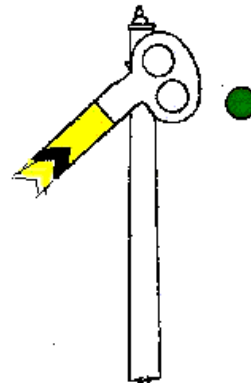
- (1) A semaphore Distant signal has a fish-tailed arm.
 (2) The aspects and indications of a semaphore Distant signal working in the lower quadrant are shown below: -

Semaphore Distant signal in Two-Aspect Signalling Territory

'On' position



'Off' position



ASPECT:

Caution

Proceed

Indication:

Proceed and be prepared to stop at the next Stop signal

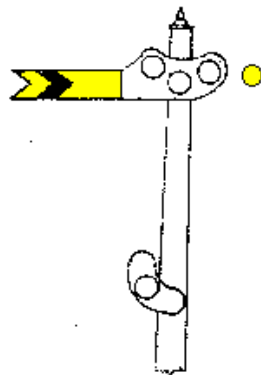
Proceed

Note: - This signal shall be provided only in Modified Lower Quadrant signalling.

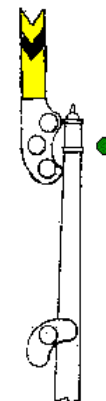
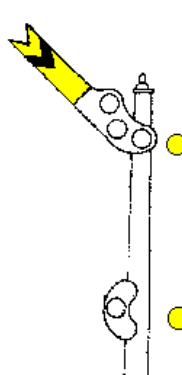
(3) The aspects and indications of a semaphore Distant signal working in the upper quadrant are shown below: -

Semaphore Distant signal in Multiple-Aspect Signalling Territory

'On' position



'Off' position



ASPECT:

Caution

Attention

Proceed

Indication:

Proceed and be prepared to stop at the next Stop signal

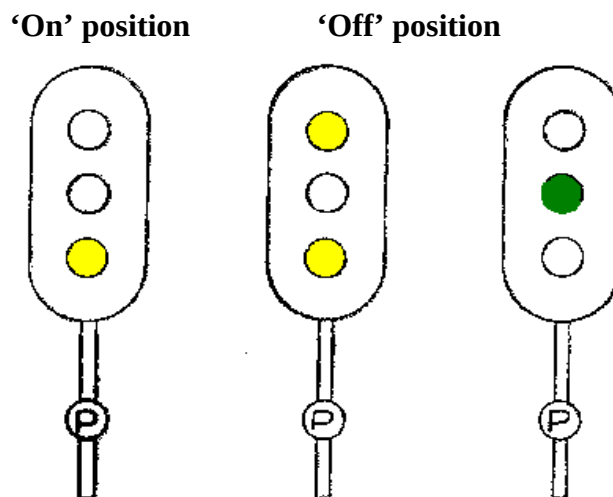
Proceed and be prepared to pass next signal at such restricted speed as may be prescribed by special Instructions. Train is being received on main line and is required to stop at the Starter signal; or on a Loop line required to stop at the Starter signal or to pass run through via Loop Line.

Proceed, Block Section ahead is clear, train is to pass run through via Main Line.

Note: - The distance between the two yellow lights shall be 1.5 metres when this signal displays 'Attention' aspect at night.

(4) The aspects and indications of a colour light Distant Signal are shown below: -

Colour light Distant signal in Multiple-Aspect Signalling Territory



ASPECT:	Caution	Attention	Proceed
Indication:	Proceed and be prepared to stop at the next Stop signal	Proceed and be prepared to pass next signal at such restricted speed as may be prescribed by special Instructions. Train is being received on main line and is required to stop at the Starter signal; or on a Loop line required to stop at the Starter signal or to pass run through via Loop Line.	Proceed, Block Section ahead is clear, train is to pass run through via Main Line.

(5) A Distant signal shall be located at an adequate distance in rear of the Stop signal, the aspect of which it pre-warns.

(6) Where necessary more than one Distant Signal may be provided. In such a case, the outermost signal, to be located at an adequate distance from the first Stop signal, shall be called the Distant Signal and the other called the Inner Distant signal, with the Distant signal capable of displaying 'Attention' or 'Proceed' aspect only.

- (7) Under approved special instructions, a colour light Distant Signal may be combined with the last Stop signal of a station in rear or with a Stop signal protecting a level-crossing.

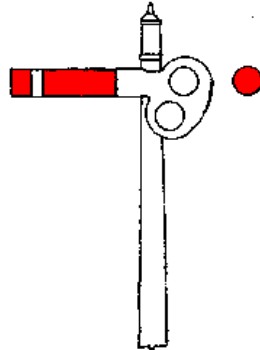
When a colour light Distant signal is combined with the last Stop signal of a station in rear or with a Stop signal protecting a level-crossing, arrangements shall be such that the signal shall not display a less restrictive aspect than the 'Stop' aspect till Line Clear had been obtained from the station ahead in the former case and until the level-crossing gates have been closed and locked for the passage of trains in the latter case.

3.08 DESCRIPTION OF STOP SIGNALS AND THEIR INDICATIONS.—

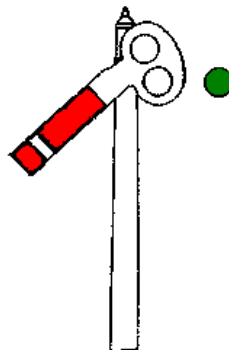
- (1) A semaphore Stop signal has a square ended arm.
- (2) The aspects and indications of a semaphore Stop signal, working in the lower quadrant are shown below: -

Semaphore Stop signal in Two-Aspect Signalling Territory

'On' position



'Off' position



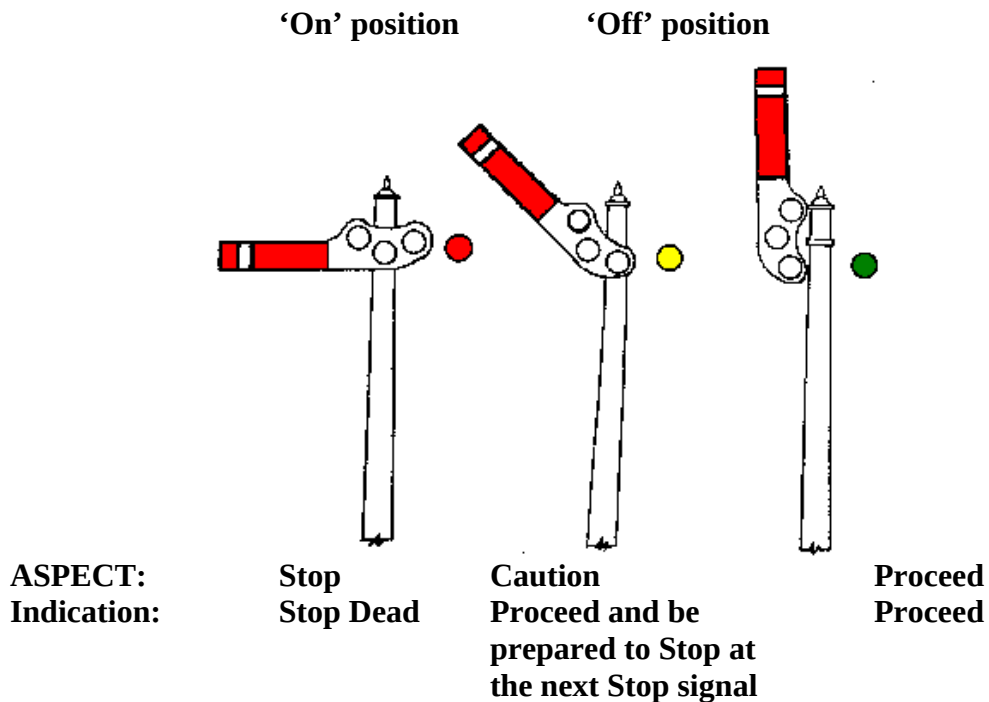
ASPECT:
Indication:

Stop
Stop dead

Proceed
Proceed

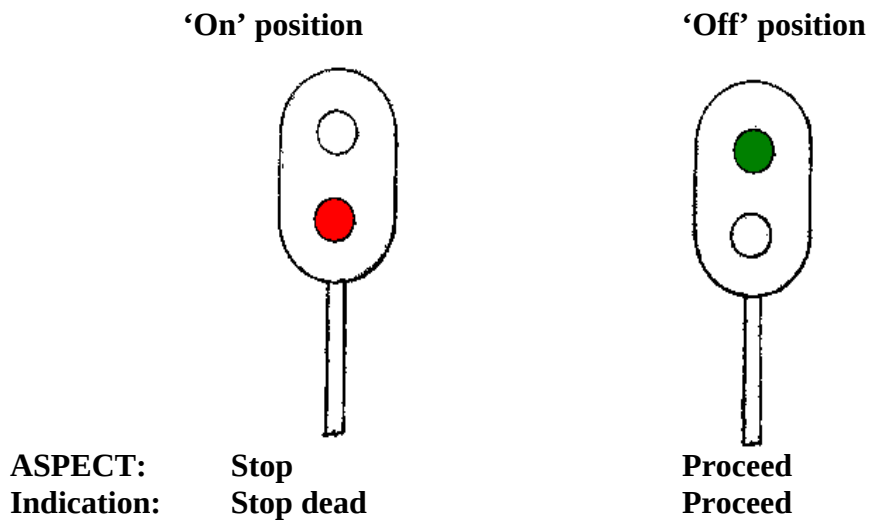
- (3) The aspects and indications of a semaphore Stop signal working in the upper quadrant are shown below: -

Semaphore Stop signal in Multiple-Aspect Signalling Territory

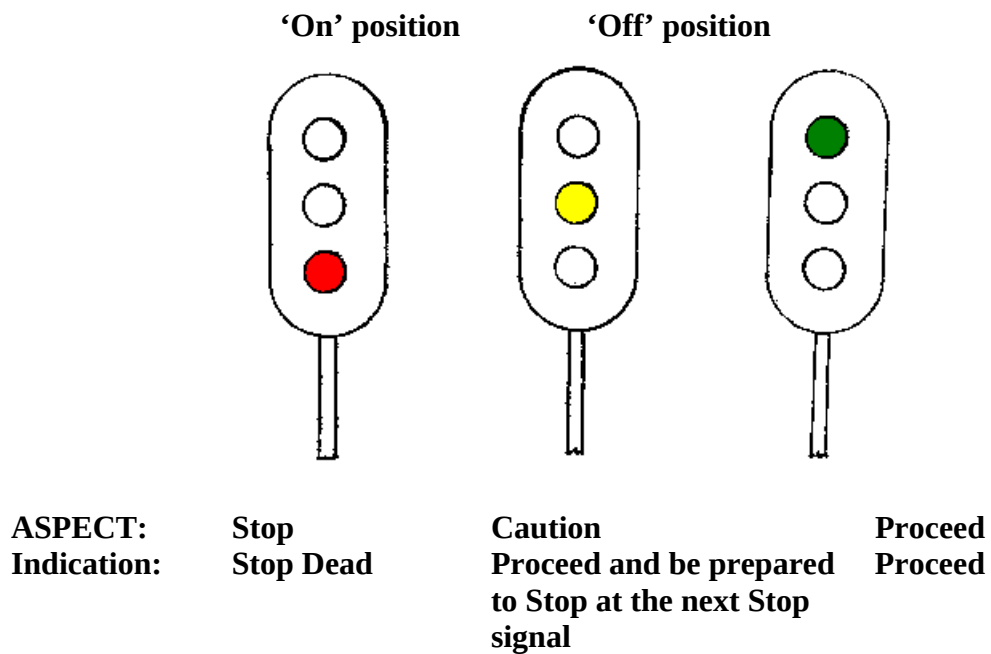


(4) The aspects and indications of a colour light Stop signal are shown below: -

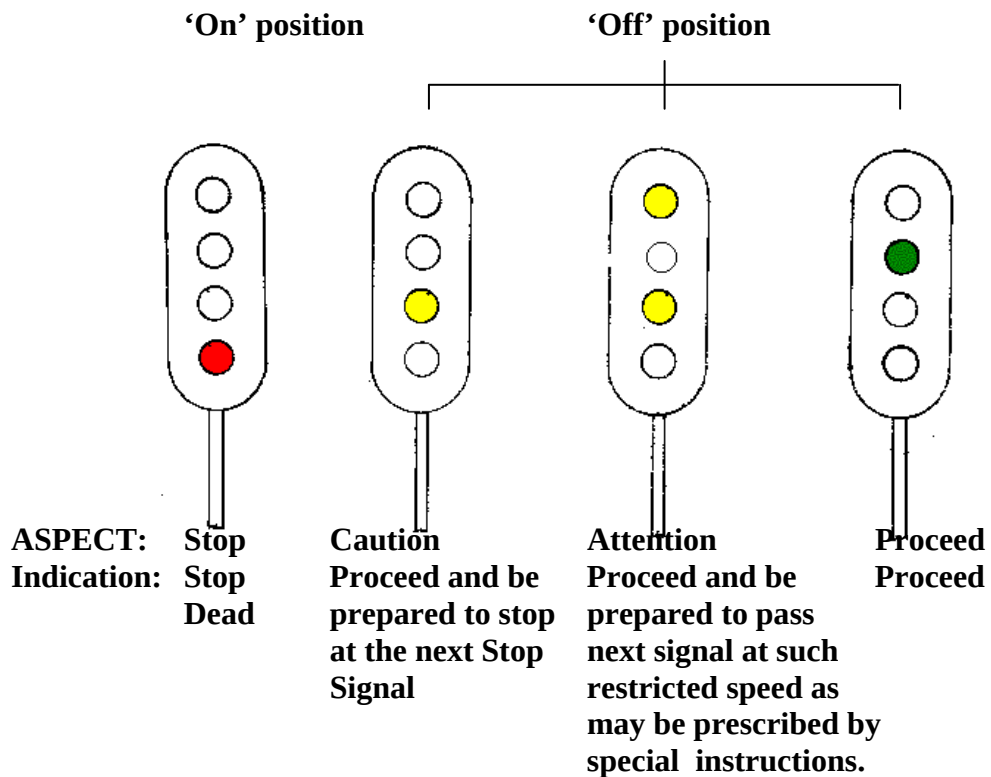
(a) Colour light Stop signal in Two-Aspect Signalling Territory



(b) Colour light Stop signal in Multiple Three-Aspect Signalling Territory



(c) Colour light Stop signal in Multiple Four—Aspect Signalling Territory



3.09 KINDS OF FIXED STOP SIGNALS FOR APPROACHING TRAINS.—

- (1) The Stop signals which control the movement of trains approaching a station are of three kinds, namely-Outer, Home and routing signals.**
- (2) The Outer Signal, where provided, is the first Stop Signal of a station and is located at an adequate distance outside the point up to which the line may be obstructed, after Line Clear has been granted to, or obtained by the station in rear.**
- (3) The Home signal is the first Stop signal of a station at which an Outer signal is not provided and the second Stop signal of a station at which an Outer signal is provided. It shall be located outside all connections on the line to which it refers.**
- (4) The Routing signal is a signal used to indicate to a Loco Pilot which of two or more diverging routes is set for him, when the Home signal is, in consequence of its position, inconvenient for this purpose.**

3.10 KINDS OF FIXED STOP SIGNALS FOR DEPARTING TRAINS.—

- (1) The Stop signals which control the movement of trains leaving a station are of two kinds, namely-Starter and Advanced Starter.**
- (2) When a train leaving a station is guided by only one starting signal, it is the last Stop signal of the station and is called the Starter.**
- (3) When a train leaving a station is guided by more than one Starter Signal, the outermost starting signal is the last Stop Signal of the station and is called the Advanced Starter.**
- (4) The Starter, where only one such signal is provided, or the Advanced Starter, shall be fixed at the limit, beyond which no train may pass, unless the Loco Pilot is given the authority to proceed required under the system of working and shall be placed outside all connections of the line, to which it refers except where otherwise allowed by approved special**

instructions. Shunting operations beyond this limit shall be carried out only in accordance with special instructions.

- (5) Where an Advanced Starter is provided, the Starter referring to any line shall be placed so as to protect the first facing points or fouling mark of the connections, to another running line.

3.11 INTERMEDIATE BLOCK STOP SIGNAL.—

Intermediate Block Stop signal is the Home signal provided at an Intermediate Block Post.

3.12 KINDS OF FIXED STOP SIGNALS IN AUTOMATIC BLOCK TERRITORIES.—

- (1) Stop signals in Automatic Block territory shall be colour light signals and may be of the following kinds-

- (a) an Automatic Stop signal which is not dependent upon manual operation but is controlled automatically by the passage of a train into, through and out of the automatic block signalling sections;

- (b) a Semi-Automatic Stop signal which is capable of being operated either as an Automatic Stop signal or as a Manual Stop signal, as required;

- (i) when a Semi-Automatic Stop signal works as an Automatic Stop signal, it assumes 'on' and 'off' aspects automatically according to the condition of the automatic block signalling sections ahead;

- (ii) when a Semi-Automatic Stop signal works as a Manual Stop signal, it assumes 'on' aspect automatically on the occupation of the automatic block signalling section ahead, but assumes 'off' aspect when operated manually, provided the relevant automatic block signalling sections ahead are clear;

- (iii) when a Semi-Automatic Stop signal works as an Automatic Stop signal, the 'A' marker provided under the signal is illuminated. When the 'A' marker is extinguished, the signal shall be deemed to work as a Manual Stop signal;

- (ba) a Modified Semi-Automatic Stop signal by converting one of the Automatic stop signal in mid-section under special instructions; when the 'A' marker is illuminated the signal works as Automatic stop signal, and when the 'A' marker is extinguished

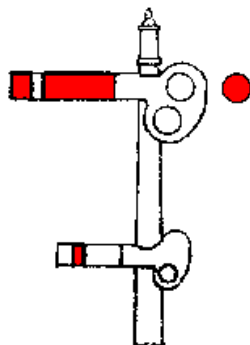
it works as modified Semi-Automatic stop signal and assumes 'off' aspect automatically or is taken 'off' manually as required; and

- (c) A Manual Stop signal operated manually and which cannot work as an Automatic or a Semi-Automatic Stop signal.
- (2) Colour light signals in automatic Block territory shall be three-aspect or four-aspect.

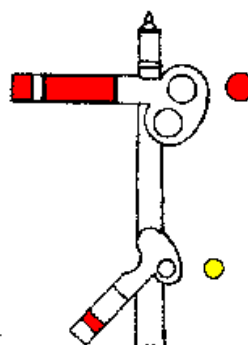
3.13 CALLING-ON SIGNALS.—

- (1) A Calling-on signal is a subsidiary signal, which has no independent aspect in the 'on' position and shall be—
 - (a) a short square ended semaphore arm, or
 - (b) a miniature colour light provided with a 'C' marker.
- (2) A Calling-on signal, where provided, shall be fixed below a Stop signal governing the approach of a train. Under approved special instructions, a calling-on signal may be provided below any other Stop signal except the last Stop signal.
- (3) A Calling-on signal, when taken 'off' calls on the Loco Pilot of a train to draw ahead with caution, after the train has been brought to a stop, even though the Stop signal above it is at 'on' and indicates to the Loco Pilot that he should be prepared to stop short of any obstruction.
- (4) A Calling-on signal shall show no light in the 'on' position.
- (5) The aspects and indications of a semaphore Calling-on signal are shown below: -
 - (a) Miniature Semaphore Arm type Calling-on signal in Two-Aspect Signalling Territory

'On' position



'Off' position



ASPECT:

Proceed slow

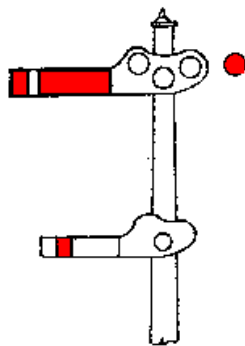
Indication:

Loco Pilot shall obey the aspect of the Stop signal

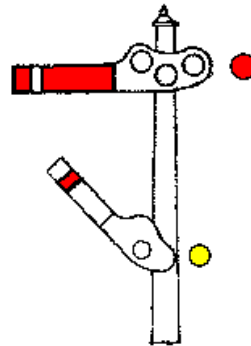
Stop and then draw ahead with caution and be prepared to stop short of any obstruction

- (b) Miniature Semaphore Arm type Calling-on signal in Multiple-Aspect Signalling Territory**

‘On’ position



‘Off’ position



ASPECT:

Proceed slow

Indication:

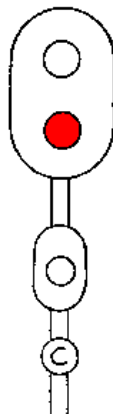
Loco Pilot shall obey the aspect of the Stop signal

Stop and then draw ahead with caution and be prepared to stop short of any obstruction

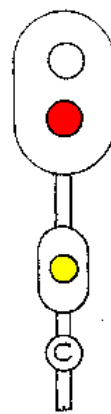
- (6) The aspects and indications of a Colour light type Calling-on signal are shown below: -**

- (a) Colour light type Calling-on signal in Two Aspect Signalling Territory**

On’ position



‘Off’ position



ASPECT:

Proceed slow

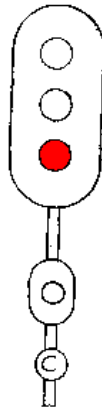
Indication:

Loco Pilot shall obey the aspect of the Stop signal

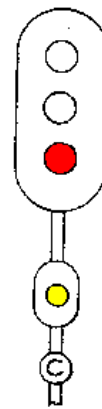
Stop and then draw ahead with caution and be prepared to stop short of any obstruction

- (b) Colour light type Calling-on signal in Multiple-Aspect Signalling Territory**

‘On’ position



‘Off’ position



ASPECT:

Proceed slow

Indication:

Loco Pilot shall obey the aspect of the Stop signal

Stop and then draw ahead with caution and be prepared to stop short of any obstruction

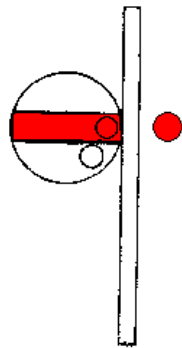
3.14 SHUNT SIGNALS.—

- (1)(a) A Shunt signal is subsidiary signal and shall be either—**
- (i) a white disc with a red bar across it, or**
 - (ii) a position light signal.**
- (b) Under special instructions, a Shunt signal may be a miniature semaphore arm.**
- (2) Shunt signals control shunting movements.**
- (3) A Shunt signal may be placed on a post by it self or below a Stop signal other than the first Stop signal of a station.**

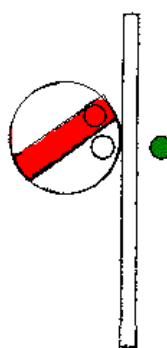
- (4) More than one Shunt signal may be placed on the same post and when so placed the topmost Shunt signal shall apply to the extreme left hand line and the second Shunt signal from the top shall apply to the next line from the left and so on.
- (5) When a Shunt signal is taken 'off', it authorises the Loco Pilot to draw ahead with caution for shunting purposes although Stop signal, if any, above it is at 'on'.
- (6) When a Shunt signal is placed below a Stop signal, it shall show no light in the 'on' position.
- (7) In case Shunt signals are not provided, hand signals may be used for shunting.
- (8) The aspects and indications of a disc type Shunt signal are shown below: -

(a) Disc type Shunt signal in Two-Aspect Signalling Territory

'On' position



'Off' position

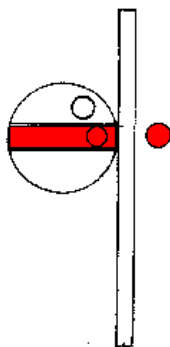


ASPECT: Stop
Indication: Stop dead

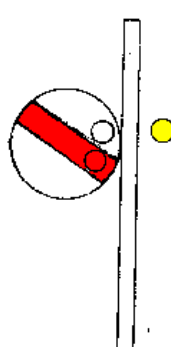
Proceed slow
Proceed with caution
for shunting

(b) Disc type Shunt signal in Multiple-Aspect Signalling Territory

'On' position



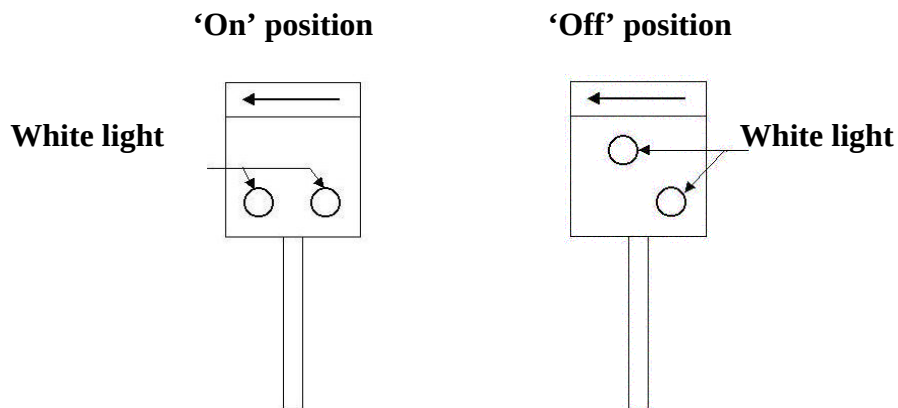
'Off' position



ASPECT:	Stop	Proceed slow
Indication:	Stop dead	Proceed with caution for shunting

- (9) The aspects and indications of a position light type Shunt signal are shown below: -

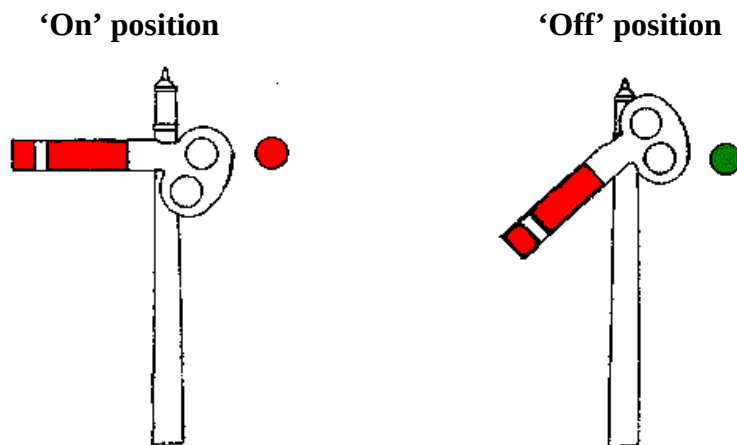
Position light type Shunt signal in Two – Aspect or Multiple – Aspect Signalling Territory



ASPECT:	Stop	Proceed slow
Indication:	Stop dead	Proceed with caution for shunting

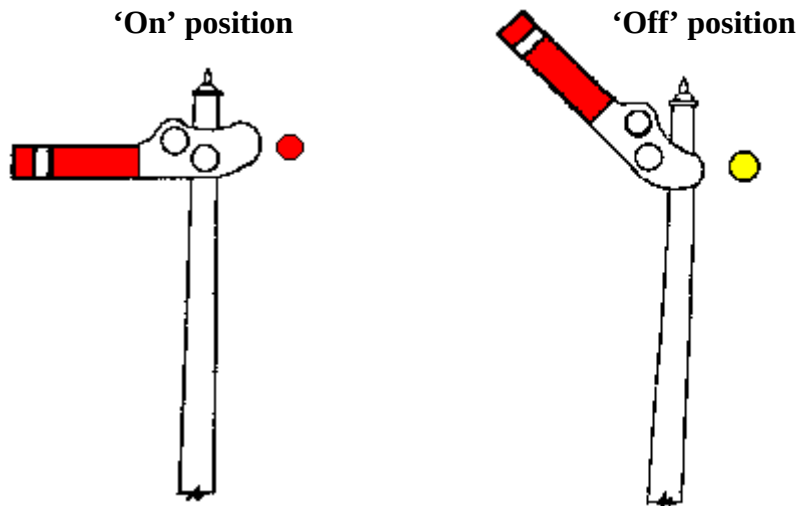
- (10) The aspects and indications of a semaphore arm type Shunt signal are shown below: -

- (a) Miniature Semaphore Arm type Shunt signal in Two Aspect signalling Territory



ASPECT:	Stop	Proceed slow
Indication:	Stop dead	Proceed with caution for shunting

(b) **Miniature Semaphore Arm type Shunt signal in Multiple-Aspect Signalling Territory**



ASPECT:	Stop	Proceed slow
Indication:	Stop dead	Proceed with caution for shunting

3.15 CO-ACTING SIGNALS.—

- (1) Co-acting signals are duplicate signals fixed below ordinary signals and are provided where, in consequence of the height of the signal post, or of there being an over-bridge or other obstacle, the main arm or light is not in view of the Loco Pilot during the whole time that he is approaching it.
- (2) Co-acting signals shall be fitted at such height that either the main arm or light, or the Co-acting arm or light, is always visible.

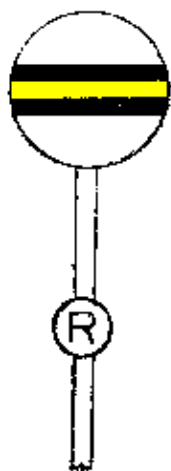
3.16 REPEATING SIGNALS.—

- (1) A Signal placed in rear of a fixed signal for the purpose of repeating to the Loco Pilot of an approaching train the aspects of the fixed signal in advance is called a Repeating Signal.

- (2) A Repeating signal shall be provided with an 'R' marker and shall be of—
- (a) banner type, or
 - (b) a square ended semaphore arm, or
 - (c) a colour light signal.
- (3) The aspects and indications of a banner type Repeating signal are shown below:

Banner type Repeating signal in Two-Aspect Signalling Territory

'On' position



Off' position



ASPECT: Signal 'On'

Indication: Signal which it repeats is at 'On'

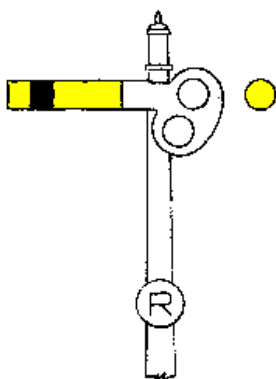
Signal 'Off'

Signal which it repeats is 'Off'

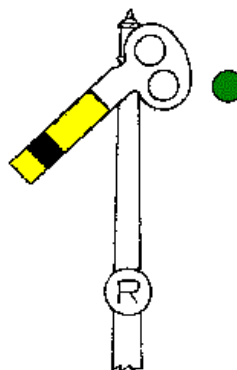
- (4) The aspects and indications of a semaphore arm type Repeating signal are shown below: -

Semaphore Arm type Repeating signal in Two-aspects Signalling Territory

'On' position



'Off' position



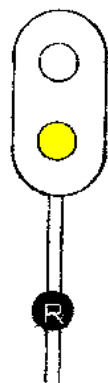
ASPECT: Signal 'On'
Indication: Signal which it repeats is at 'On'

Signal 'Off'
Signal which it repeats is 'Off'

- (5) The aspects and indications of a colour light type repeating signals are shown below: -

Colour light type Repeating signal

'On' position



'Off' position



ASPECT: Signal 'On'
Indication: Signal which it repeats is at 'On'

Signal 'Off'
Signal which it repeats is 'Off'

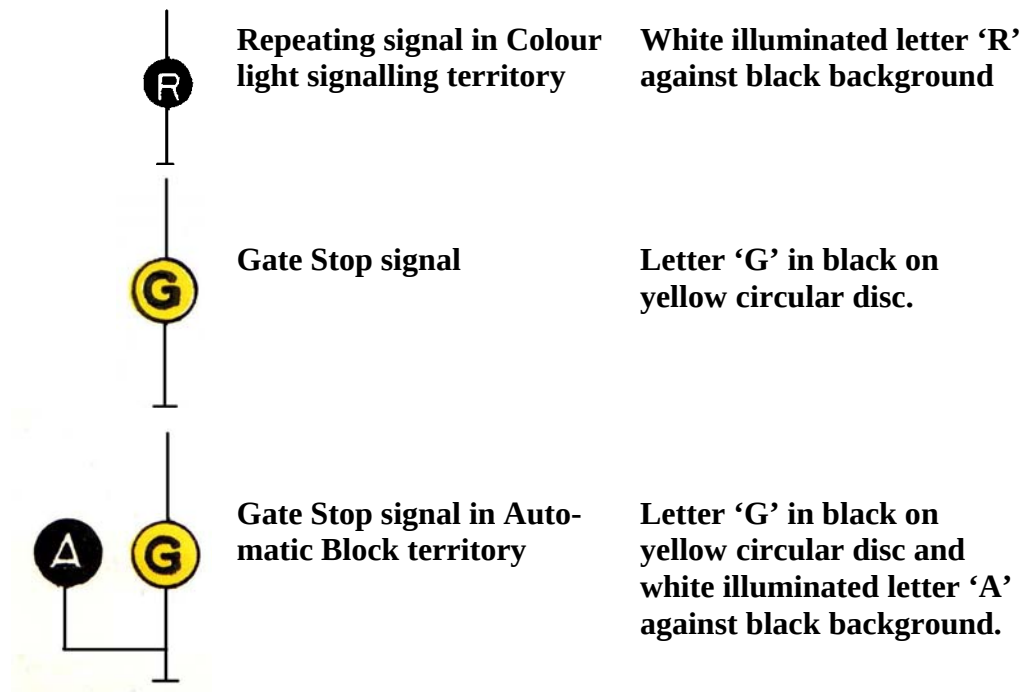
3.17 DISTINGUISHING MARKER AND SIGNS FOR SIGNALS. —

- (1) Where necessary, signals shall be distinguished by prescribed markers. Such markers shall be fixed on the signal posts below the signal as under.

<u>Appearance</u>	<u>Provided on</u>	<u>Description</u>
	Automatic Stop signal	Letter 'A' in black on white circular disc.
	Semi-Automatic Stop Signal	White illuminated letter 'A' against black background when working as an Automatic Stop signal and letter 'A' extinguished when working as a Manual Stop signal
	Colour light Distant or Warner signal on a post by itself.	Letter 'P' in black on white circular disc.

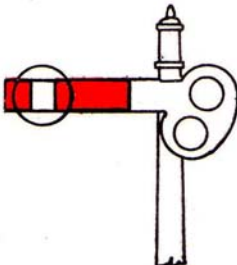
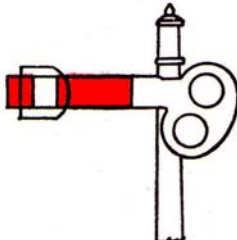
Note: - Where a colour light Distant signal is combined with a last Stop signal as provided for under sub-rule(7) of Rule 3.07, the marker shall be dispensed with.

<u>Appearance</u>	<u>Provided on</u>	<u>Description</u>
	Intermediate Block Stop signal	Letter 'IB' in black on white circular disc.
	Calling-on signal	Letter 'C' in black on white circular disc.
	Repeating signal in sema-Phore signalling territory	Letter 'R' in black on white circular disc.



Note: - Letter 'A' shall be lit only when the gates are closed and locked against road traffic.

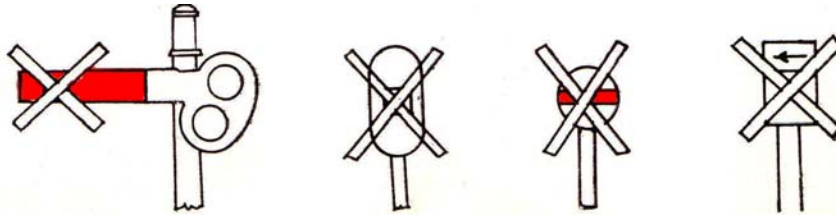
(2) Where necessary, signal arms shall be distinguished by prescribed signs as under:

Appearance	Provided on	Description
	●	Approach Stop signal for Goods running lines only.
	●	Letter 'D' in black on semaphore arm

- (3) Other distinguishing markers or signs may be used with the approval of the Board.

3.18. SIGNALS OUT OF USE.—

- (1) When a fixed signal is not in use, it shall be distinguished by two crossed bars, each bar being not less than 1 metre long and 10 centimetres wide, as illustrated below: -



- (2) A semaphore or disc signal when not in use shall be kept fixed in the 'on' position.
- (3) Signals not in use shall not be lit.

3.19 PLACING OF STOP SIGNALS AT DIVERGING JUNCTIONS.—

Unless otherwise permitted by approved special instructions, where two or more lines diverge, the signals shall be fixed on a bracket post or an approved type of route indicator shall be provided instead of separate signals;

Provided that for speed up to 75 kilometres per hour with manually operated multiple aspect signals, only a single arm Home signal may be provided instead of separate signals on a bracket post or a route indicator. The facing points must be provided with point indicators.

3.20 PLACING OF STOP SIGNALS AT CONVERGING JUNCTIONS.—Unless otherwise permitted by approved special instructions, where two or more lines converge, signals shall be placed on separate posts. Where the number of signals is considerable, these may be provided on a bracket post or a signal bridge or gantry.

3.21 SIGNALS ON BRACKET POST OR SIGNAL BRIDGE OR GANTRY.— Where signals are placed on a bracket post or a signal bridge or a gantry, these shall be

- (a)** so grouped that the respective signals are easily distinguishable for each running line and are placed as nearly as possible over the running lines to which they refer,
- (b)** so placed that the signal referring to the main lines is higher than the signal or signals referring to the other running line or lines and
- (c)** so arrange that the extreme left hand signal refers to the extreme left hand line and the second signal from the left refers to the next line from the left and so on.

3.22 PLACING OF MORE THAN ONE SIGNAL ON THE SAME POST.—

- (1)** Not more than one signal referring to trains moving in the same direction, whether on the same line or on separate lines, shall be placed on the same post, except—
 - (a)** as prescribed in these rules for Calling-on, Shunt, Co-acting and Warner signals, or
 - (b)** under approved special instructions.
- (2)** Where under approved special instructions more than one signal is placed on the same post, the topmost signal shall apply to the extreme left hand diverging line and the second signal from the top shall apply to the next line from the left and so on.

Provided that in exceptional cases, where two home signals are placed on the same post, under approved special instructions, the top signal shall apply to the main line and the lower signal shall apply to the other lines.

3.23 ELECTRIC REPEATER.—The arm and light of any fixed signal which cannot be seen from the place from which the signal is worked shall be repeated to such place by means of an efficient electric repeater.

3.24 BACK-LIGHTS.—

- (1) Every semaphore or disc signal, the light of which cannot be seen from the place from which the signal is worked, shall be provided with a back-light to indicate whether the signal light is burning or not.**
- (2) Back-light of signals shall show a small white light when ‘on’ and no light at all, in any other position.**
- (3) Any fixed light used in conjunction with a semaphore signal shall show a back light.**
- (4) Back-lights may not be provided when alternative arrangements are made at the place from which the signal is worked to indicate whether signal lights are burning or not.**

C. EQUIPMENT OF SIGNALS

3.25 OBLIGATION TO PROVIDE FIXED SIGNALS AT STATIONS.—Fixed signals prescribed in this sub-chapter shall be provided at every station, except—

- (a) at stations between which trains are worked on the One Train Only System and**
- (b) at stations which are exempted from the provision of signals under approved special instructions.**

3.26 COMMISSIONING OF FIXED SIGNALS.—Fixed signals shall not be brought into use until they have been passed by the Commissioner of Railway Safety as being sufficient to secure the safe working of trains.

3.27 MINIMUM EQUIPMENT OF FIXED SIGNALS AT STATIONS PROVIDED WITH MANUALLY OPERATED MULTIPLE-ASPECT SIGNALLING.—The minimum equipment of fixed signals to be provided for each direction shall be as follows—

- (a) at class ‘B’ stations ... a Distant, a Home and a Starter and**
- (b) at class ‘C’ stations ... a Distant and a Home.**

3.28 MINIMUM EQUIPMENT OF FIXED SIGNALS AT STATIONS PROVIDED WITH MODIFIED LOWER QUADRANT SIGNALLING.—Modified lower quadrant signalling may be introduced only where it is expressly sanctioned by a special order of the Railway Board. The minimum equipment of fixed signals to be provided for each direction shall be as follows—

- (a) at class ‘B’ stations ... a Distant, a Home, a Warner below the Main Home and a Starter and
- (b) at class ‘C’ stations ... a Distant and a Home.

3.29 MINIMUM EQUIPMENT OF FIXED SIGNALS AT OTHER STATIONS PROVIDED WITH TWO-ASPECT SIGNALLING.—The minimum equipment of fixed signals to be provided for each direction shall be as follows—

- (a) at class ‘A’ stations ... a Warner, a Home and a Starter,
- (b) at class ‘B’ stations—
 - On a single line... an Outer and a Home,
 - On a double line... an Outer, a Home and a Starter and both on a single and a double line a Warner shall be provided in accordance with Rule 3.06, if trains run through at a speed exceeding 50 kilometres an hour without stopping and
- (c) at class ‘C’ stations ... a Warner and a Home.

3.30 ADDITIONAL FIXED SIGNALS AT STATIONS GENERALLY.—In addition to the minimum equipment of signals prescribed in Rules 3.27, 3.28, 3.29 and 3.32 such other fixed signals shall be provided at every station as may be necessary for the safe working of trains.

3.31 SIGNALS AT CLASS ‘D’ STATIONS.—At a class ‘D’ station, a train may be stopped in such manner as may be authorised by special instructions.

**3.32 PROVISION OF AN ADVANCED STARTER, SHUNTING
LIMIT BOARD OR BLOCK SECTION LIMIT BOARD.—**

- (1) On a single line class ‘B’ station worked on the Absolute Block System if the obstructing of the line outside the Home signal or the outermost facing points in the direction of an approaching train is permitted under special instructions under Rule 8.09, a Shunting Limit Board or an Advanced Starter shall be placed at such shunting distance from the Home signal or the outermost facing points as local conditions may require, provided the distance between the Shunting Limit Board (bearing the words ‘Shunting Limit’ on the side which faces the station and fitted with a lamp showing a white light in both directions to mark its position by night) or the Advanced Starter and the opposing first Stop signal is never less than 400 metres in the two-aspect signalling territory and 180 metres in the multiple-aspect or modified lower quadrant signalling territory. The location of such board or Advanced Starter shall mark the limit up to which shunting may be permitted.**
- (2) On a double line class ‘B’ station worked on Absolute Block System equipped with multiple-aspect or modified lower quadrant signalling and where there are no points or the outermost points at the approaching end are trailing, a Block Section Limit Board (bearing the words ‘Block Section Limit’ on the side which faces the station and fitted with a lamp showing white light in both directions to mark its position by night) shall be provided. It shall be placed at a distance of not less than 180 metres in advance of the Home signal and shall protect the fouling mark of outermost trailing points if any. The location of such board shall mark the limit of the block section at such stations.**

**3.33 EXCEPTIONS TO RULES 3.27, 3.28, 3.29 AND 3.32-
Notwithstanding anything contained in Rules 3.27, 3.28, 3.29
and 3.32.**

- (a) if the station has only one connection ‘off’ the main line, the stations shall be worked in accordance with approved special instruction;**
- (b) on any section where traffic is light and speeds slow, one Stop signal only in each direction may be provided at each station, such signal to be located at an adequate distance outside the outer most facing points of the station and trains worked in accordance with approved special instructions and**

- (c) on any Railway having very light traffic, all signals may be dispensed with and trains worked under approved special instructions;

Provided that at stations with manually operated multiple aspect signals, where the speed of trains through a station does not exceed 50 kilometres per hour, a Distant signal and Home signal only may be provided in each direction under approved special instructions.

3.34 FIXED SIGNALS AT LEVEL CROSSINGS.—

- (1) Unless exempted under approved special instructions, every level-crossing gate which closes across the line at a level-crossing shall, except when interlocked with station signals, be provided with signals fixed at an adequate distance from the level-crossing showing Stop aspects in both Up and Down directions, when the gates are open for the passage of road traffic.
- (2) Except where otherwise prohibited under special instructions, a 'G' marker shall be provided on a gate Stop signal.

3.35 PROTECTION AND WORKING OF POINTS OF OUTLYING SIDINGS.—Where there are points in the main line at a place which is not a block station, provision for the protection of such points, by signals or otherwise and for working them, shall be made in order to secure the safe working of trains, as laid down under approved special instructions.

D. WORKING OF SIGNALS AND POINTS

3.36 FIXED SIGNALS GENERALLY.—

- (1) Every fixed signal shall be so constructed that, in case of failure of any part of its connections, it shall remain at, or return to its most restrictive aspect.
- (2) A signal which has been taken 'off' for the passage of a train shall not be placed 'on' until the whole of the train which it controls has passed it, except-

- (a) In case of emergency, or

(aa) where Starter and Advanced Starter taken 'off' for departing trains that is trains starting from station after coming to stop are required to be put back for the purpose of movement of another train for precedence or crossing shall be put back only after taking following precautions:-

(i) relevant Starter and Advanced Starter may be replaced to 'on' position and thereafter the Loco Pilot of the train for which the signals had been taken 'off' shall be advised by on-duty Station Master through a secured means of communication, specified under special instructions or where secured means of communication are not available, through a written memo to the effect that the said signals have been replaced to 'on' and that the Loco Pilot shall not start;

(ii) till the Loco Pilot has been advised through secured means of communication referred to in sub-clause (i) or through a written memo and his acknowledgment received, the route set shall not be altered except to avert an accident; or

(b) Where arrangement is provided to restore the signal to 'on' automatically, the control operating the signal shall not be restored to its normal position till the whole of the train has passed it.

(3) No fixed signal within station limits shall be taken 'off' without the permission of the Station Master and in the case of a signal outside the station limits without the permission of such person as may for the time being be in independent charge of the working of such signal.

3.37 NORMAL ASPECT OF SIGNALS.—

(1) Unless otherwise authorised under approved special instructions, fixed signals except automatic signals, shall always show their most restrictive aspect in their normal position.

(2) The normal aspect of an Automatic Stop signal is 'Proceed'. Where, however, the signal ahead is manually operated the aspect normally displayed may be 'Caution' or 'Attention'.

3.38 POINTS AFFECTING MOVEMENT OF TRAIN.—

(1) The Station Master shall not give permission to take signals 'Off' for a train until –

- (a) All facing points over which the train will pass are correctly set and locked.
 - (b) All trailing points over which the train will pass are correctly set and
 - (c) The line over which the train is to pass is clear and free from obstructions.
- (2) When a running line is blocked by a stabled load, wagon vehicle or by a train which is to cross or give precedence to another train or immediately after the arrival of a train at the station etc. the points in rear on double line sections and at either end on single line sections should be immediately set against the blocked line except when shunting or any other movement is required to be done immediately in that direction on that line.

3.39 LOCKING OF FACING POINTS.—Facing points, when neither interlocked nor key locked, shall be locked for the passage of a train either by a clamp, or by a through bolt, with a padlock. It is not sufficient to lock the lever working the points.

3.40 CONDITIONS FOR TAKING 'OFF' HOME SIGNAL.—

- (1) When a train is approaching a Home Signal otherwise than at a terminal station, the signal shall not be taken 'off' until the train has first been brought to a stand outside it, unless.—
 - (a) on a double line, the line is clear for an adequate distance beyond the Starter; or
 - (b) on a single line, the line is clear for an adequate distance beyond the trailing points, or for an adequate distance beyond the place at which the train is required to come to a stand.”
- (2) Where a train has first been brought to a stand outside the Home signal, the signal may be taken 'off', if.—
 - (a) On a double line, the line is clear up to the Starter, or
 - (b) On a single line, the line is clear up to the trailing points or under approved special instructions up to the place at which the train is required to come to a stand.

- (3) Except under approved special instructions, the adequate distance referred to in sub-rule (1) shall never be less than—
- (a) 180 metres at stations equipped with two aspect lower quadrant or two aspect colour light signals, or
 - (b) 120 metres in the case of stations provided with multiple-aspect signals or modified lower quadrant signals.
- (4) Where a sand hump of approved design, or under approved special instructions a derailing switch, has been provided for the line on which a train is to be received, they shall be deemed to be efficient substitutes for the adequate distance referred to in subrule (3).

3.41 CONDITIONS FOR TAKING 'OFF' OUTER SIGNAL.—

- (1) When a train is approaching the outer signal otherwise than at a terminal station, the signal shall not be taken 'off', until the train has first been brought to a stand outside the signal, unless the line on which the train is to be received in the station is clear—
- (a) in the case of a double line, up to the starter signal and
 - (b) in the case of single line, for an adequate distance beyond the first facing points.
- (2) where the train has first been brought to a stand outside the outer signal, the signal shall not be taken 'off', unless the line is clear up to the first facing points, or up to the Home signal at a station, where there are no facing points.

3.42 CONDITIONS FOR TAKING 'OFF' LAST STOP SIGNAL OR INTERMEDIATE BLOCK STOP SIGNAL. (1) On double line, the last stop signal or Intermediate Block Stop signal shall not be taken 'off' for a train unless Line clear has been obtained from the block station in advance.

- (2) On single line-
- (a) the last stop signal shall not be taken 'off' for a train unless line clear has been obtained from the block station in advance;
 - (b) for Intermediate block signaling-
 - (i) first, the direction of traffic shall be established and then line clear shall be obtained from the block station in advance as per the established direction of traffic'
 - (ii) only after establishing the direction of traffic the train movement in the 'Station controlled Intermediate Block section' shall be permitted; and
 - (iii) the Intermediate Block Stop Signal shall not be taken 'off' unless the line clear has been obtained from block station in advance and direction of traffic is established.

Explanation.-On Single Line Intermediate Block signaling, the line between adjacent block station is divided into two subsections, the first section which shall be termed as 'station controlled intermediate block section' and the section between Intermediate Block signal to First Stop Signal of block station ahead shall be termed as 'block controlled Intermediate Block section'.

3.43 CONDITIONS FOR TAKING 'OFF' WARNER SIGNAL. —A Warner signal shall not be taken 'off' for a train that is booked to stop or for a train that has to be stopped out of course.

3.44 CONDITIONS FOR TAKING 'OFF' GATE STOP SIGNAL. — A gate Stop signal shall not be taken 'off' until the concerned

level-crossing or crossings is or are free from obstruction and the gates of such level-crossing or crossings are closed and locked against road traffic. Where a gate Stop signal is interlocked with station signals, it shall be worked in accordance with special instructions.

3.45 CONDITIONS FOR TAKING ‘OFF’ CALLING-ON SIGNAL.-
A calling-on signal shall not be taken ‘off’ until the train has been brought to a stand at the Stop signal below which the calling-on signal is provided.

3.46 USE OF FIXED SIGNALS FOR SHUNTING.—

- (1) The Outer, Home and last Stop signal of a station shall not be taken ‘off’ for shunting purposes.**
- (2) At stations where Advanced Starter are provided, starters may be taken ‘off’ for shunting purposes, except where the interlocking interferes with this practice, in which case hand signal shall be used, where shunting signals are not provided.**

3.47 TAKING ‘OFF’ SIGNALS FOR MORE THAN ONE TRAIN AT A TIME.—When two or more trains are approaching simultaneously from any direction, the signals for one train only shall be taken ‘off’, other necessary signals being kept at ‘on’ until the train for which the signals have been taken ‘off’ has come to a stand at the station, or has cleared the station and the signals so taken ‘off’ for the said train have been put back to ‘on’ except where under special instructions, the interlocking or the layout of the yard renders a contrary procedure safe.

3.48 STOPPAGE OF TRAINS OUT OF COURSE AT STATIONS PROVIDED WITH TWO- ASPECT SIGNALLING.—When a train which is booked to run through has to be stopped out of course at a station equipped with two-aspect signals, it shall not be received until—

- (a) at stations provided with working Warners but not provided with Starters, the working Warner is kept at ‘on’;**
- (b) at stations provided with Starters but not provided with working Warners, the relevant Starters is kept at ‘on’;**
- (c) at stations provided with both working Warners and Starters, both the signals are kept at ‘on’; and**

- (d) at stations provided with neither a working Warner nor a Starter, the first Stop signal is kept at 'on' and the train brought to a stand outside it.

3.49 CARE AND LIGHTING OF SIGNAL LAMPS. —

- (1) The Station Master shall see that lamps of fixed signals, indicators and boards such as shunting Limit Board, Block Section Limit Board and Stop Board at his station are lighted at sunset and are not put out until after sunrise, or at such earlier or later time as may be prescribed by special instructions.
- (2) Sub-rule (1) shall not apply to—
 - (a) approach lighted signals,
 - (b) colour light and position light signals which shall be kept lit throughout the day and night and
 - (c) the sections where no train is scheduled to run at night.
- (3) The Station Master shall ensure that the lamps of fixed signals, indicators and boards such as Shunting Limit Board, Block Section Limit Board and Stop Board, when lit, are burning brightly and that the lenses of lamps and spectacle glasses are properly cleaned and back-lights clearly visible.
- (4) Whenever night signals are used, the Station Master shall not grant Line Clear unless he has ensured, either personally or in the manner prescribed under special instructions that the lamps of fixed signals at his Station which are not approach lighted and which apply to the train are burning. If signal lights cannot be kept burning he shall, before giving Line Clear, initiate action in accordance with the procedure prescribed in Rules 3.68 to 3.72.
- (5) Before lighting a semaphore signal or indicator lamp, the railway servant deputed for lighting it, shall inspect the lenses and spectacle glasses. In case he finds the red roundel broken, cracked or missing, he shall not light the lamp and shall report the fact immediately to the Station Master who shall treat the signal as defective.
- (6) Every railway servant in charge of signals shall see that the greatest care is taken in the focusing, cleaning and trimming of signal lamps.

3.50 TRAPS, SLIP SIDINGS AND CATCH SIDINGS.—The Station Master shall take steps to ensure that the points of all traps, slip sidings and catch sidings and other points are set against the line which they are intended to isolate, except when it is not necessary that they should be open for the purpose of isolation.

3.51 POINTS. —

- (1) All points shall normally be set for the straight except when otherwise authorised by special instructions.**
- (2) The railway servant concerned with the operation of points and signals shall not, while on duty, leave the place of operation of points or signals which are under his charge except under special instructions.**
- (3) No railway servant shall interfere with any points, signals or their fittings, signal wires or any interlocking or block gear for the purpose of effecting repairs, or for any other purpose, except with the previous permission of the Station Master.**

E. HAND SIGNALS

3.52 EXHIBITION OF HAND SIGNALS.—

- (1) All hand signals shall be exhibited by day by showing a flag or hand and by night by showing a light as prescribed in these rules.**
- (2) During day a flag or flags shall normally be used as hand signal. Hand shall be used in emergencies only when flags are not available.**
- (3) During night a hand signal shall normally be given by showing a red or green light. A white light waved violently shall be used as a Stop signal only when the red light is not available.**
- (4) Red or green light referred to in sub-rule (3) shall be either a static or flashing type.**

3.53

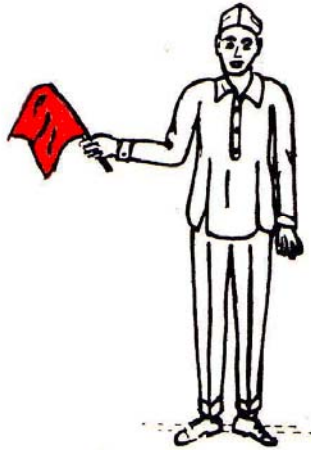
STOP HAND SIGNAL. —

Indication:

Stop Dead

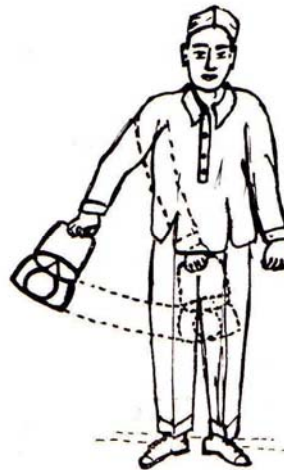
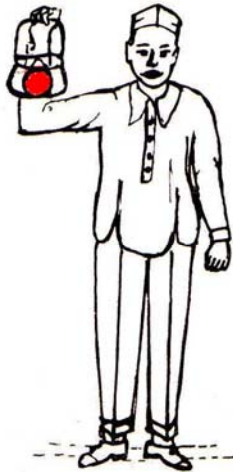
HOW GIVEN BY DAY:

By showing a red flag or by raising both arms with hands above the head as illustrated below:



HOW GIVEN BY NIGHT:

By showing a red light or by violently waving a white light horizontally across the body of the person showing the signal as illustrated below:



3.54. PROCEED HAND SIGNAL. —

Indication:

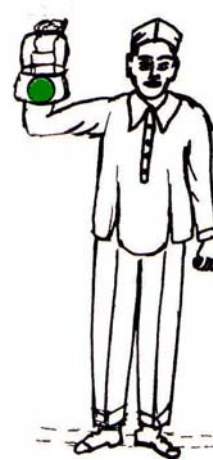
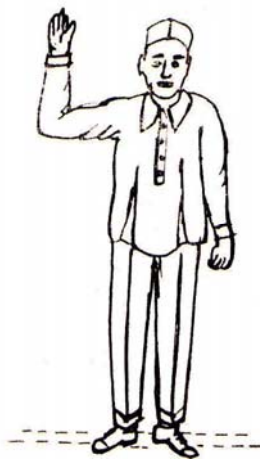
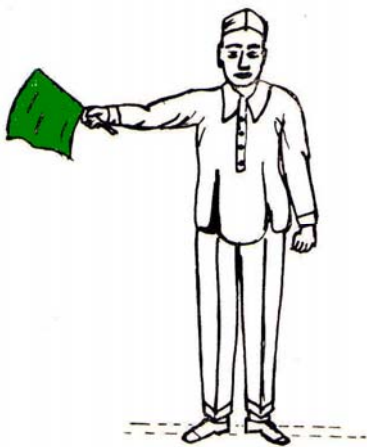
Proceed

HOW GIVEN BY DAY:

By holding a green flag or by holding one arm steadily as illustrated below:

HOW GIVEN BY NIGHT:

By holding a green light steadily as illustrated below:



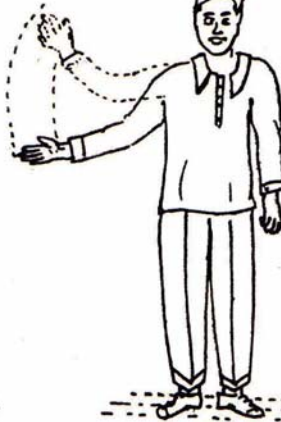
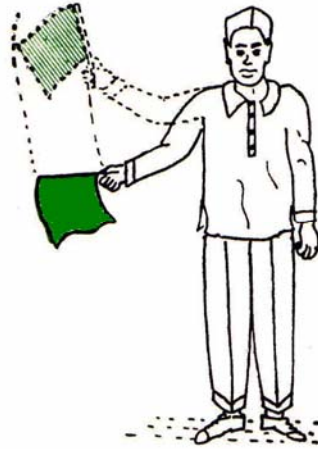
3.55

PROCEED WITH CAUTION HAND SIGNAL.—

Indication: Proceed slowly reducing speed further if the signal is given at a progressively slower rate.

HOW GIVEN BY DAY:

By waving a green flag vertically up and down or by waving one arm in a similar manner as illustrated below:



HOW GIVEN BY NIGHT:

By waving a green light vertically up and down as illustrated below:



Note: -When the speed is to be reduced further, this signal shall be given at a slower and slower rate and when a stop is desired, the Stop hand signal shall be shown.

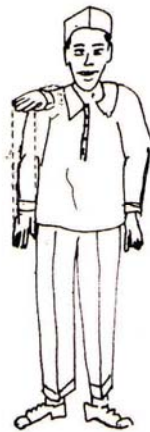
3.56

HAND SIGNALS FOR SHUNTING.—The following hand signals shall be used in shunting operations in addition to the stop hand signals.

(a) **Indication:** Move away from the person signalling.

HOW GIVEN BY DAY:

By a green flag or one arm moved slowly up and down as illustrated below:



HOW GIVEN BY NIGHT:

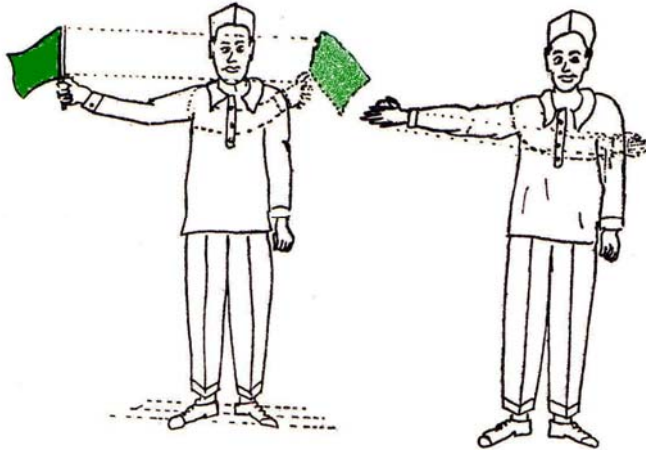
By a green light moved slowly up and down as illustrated below:



(b) Indication: Move towards the person signalling.

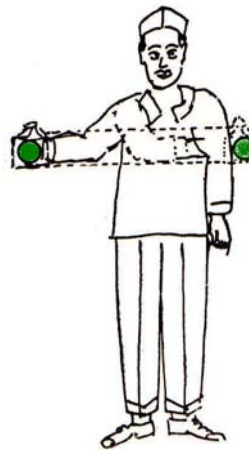
HOW GIVEN BY DAY:

By a green flag or one arm moved from side to side across the body as illustrated below:



HOW GIVEN BY NIGHT:

By a green light moved from side to side across the body as illustrated below:



Note: -The hand signals for 'Move away from the person signalling' and 'Move towards the person signalling' shall be displayed slower and slower, until the Stop hand signal is given if it is desired to stop.

(c) Indication: Move slowly for coupling.

HOW GIVEN BY DAY:

By a green and a red flag held above the head or both hands raised over the head and moved towards and away from each other as illustrated thus:



HOW GIVEN BY NIGHT:

By a green light held above the head and moved by twisting the wrist as illustrated thus:



3.57 BANNER FLAGS.- A banner flag is a temporary fixed danger signal consisting of a red cloth supported at each end on a post and stretched across the line to which it refers.

3.58 KNOWLEDGE AND POSSESSION OF HAND SIGNALS.—

- (1)** Every railway servant connected with the movement of trains, shunting operations, maintenance of installations and works of any nature affecting safety of trains shall have-

 - (a)** a correct knowledge of hand signals; and
 - (b)** the requisite hand signals with him while on duty and keep them in good working order and ready for immediate use.
- (2)** Every railway servant shall see that the staff under him concerned with use of hand signals are adequately supplied with all necessary equipment for hand signalling and have a correct knowledge of their use.
- (3)** A red flag and a green flag by day or a lamp, which is capable of showing red, green and white light by night, shall constitute the requisite equipment for hand signalling.
- (4)** Every Station Master shall see that his station is adequately supplied with all necessary equipment for hand signalling.

F. DETONATING SIGNALS

3.59 DESCRIPTION OF DETONATING SIGNALS.—Detonating signals, otherwise known as detonators or fog signals, are appliances which are fixed on the rails and when an engine or a vehicle passes over them, they explode with a loud report so as to attract the attention of the Loco Pilot.

3.60 METHOD OF USING DETONATORS.—

- (1)** A detonator when required to be used shall be placed on the rail with the label or brand facing upwards and shall be fixed to the rail by bending the clasps around the head of the rail.
- (2)** In case of a mixed gauge, detonators shall be placed on the common rail or one rail of each gauge.

3.61 PLACING OF DETONATORS IN THICK, FOGGY OR TEMPESTUOUS WEATHER IMPAIRING VISIBILITY.—

- (1)** In thick, foggy or tempestuous weather impairing visibility, whenever it is necessary to indicate to the Loco Pilot of an approaching train the locality of a signal, two detonators shall be placed on the line by a railway servant appointed by the Station Master in this behalf, about 10 metres apart and at least 270 metres outside the signal or signals concerned.
- (2)(a)** The Station Master may comply with the provision of sub-rule (1) at his discretion; but shall always do so when visibility conditions from any cause prevent him from seeing a prescribed visibility test object from a distance of not less than 180 metres or a lesser distance if expressly sanctioned by the Railway Board.
- (b)** The visibility test object may be—

 - (i)** a post erected for the purpose and lighted at night; or
 - (ii)** the arm by day and light or the back-light by night of a fixed semaphore signal specified by special instructions; or
 - (iii)** the light of a fixed colour light signal both by day and night specified by special instructions.

3.62 PLACING OF DETONATORS IN CASE OF OBSTRUCTION.—

- (1)** Whenever in consequence of an obstruction of a line, it is necessary for a railway servant to stop approaching trains, he shall proceed, plainly showing his Stop hand signal, to a point 400 metres from the obstruction and place on the line one detonator and then proceed to a point 800 metres from the obstruction and place on the line three detonators, about 10 metres apart, at such place:
Provided that on the broad gauge the first detonator shall be placed at 600 metres and three detonators at 1200 metres from the obstruction about 10 metres apart from each other.
- (2)** If the said railway servant is recalled before the obstruction is removed, he shall leave down three detonators and, on his way back, pick up the intermediate detonator.

- 3.63 REPLACEMENT OF DETONATORS ON THE LINE.—**Every railway servant placing detonators on the line shall see that they are, when necessary, replaced immediately after a train has passed over them.
- 3.64 KNOWLEDGE AND POSSESSION OF DETONATORS.—**
- (1)(a)** All Station Master, Guards, Loco Pilots, Gangmates, Gatemen and all other railway servants on whom this duty is laid by the Railway Administration, shall keep a stock of detonators.
 - (b)** The Railway Administration shall be responsible for the supply, renewal, periodical testing and safe custody of such detonators and for ensuring that their use is properly understood.
 - (2)** Every railway servant concerned with the use of detonators shall have a correct knowledge of their use and keep them ready for immediate use.
 - (3)** Every railway servant shall see that the railway servants in his charge concerned with the use of detonators have a correct knowledge of their use.

G. Signals to warn incoming train of danger ahead:

- 3.65.** The signals to be used to warn the incoming train of an obstruction shall be a red flashing hand signal lamp at night or a red flag during day.
- 3.66.** Use of warning signals.- When it becomes necessary to protect an obstruction in a block section, a signal may be used, as prescribed by special instructions under rule 3.65, while the railway servant proceeds to place detonators.
- 3.67. Knowledge and possession of warning signals. -** (1)(a) All concerned railway servant on whom this duty is laid by the Railway Administration shall keep a stock of such signal as may be prescribed by special instructions under rule 3.65;
- (b)** the Railway Administration shall be responsible for the supply, renewal and safe custody of such signals as may be prescribed by special instructions under rule 3.65 and for ensuring that their use is properly understood;

- (c) The Railway Administration shall supply every Guard, Loco Pilot, Patrolman and Gateman working on the Double or Multiple line, Ghat, Suburban or Automatic Block Territories with such signal as may be prescribed by special instructions under rule 3.65.
- (2) Every railway servant concerned with the use of signals as prescribed by special instructions under rule 3.65, shall have a correct knowledge of their use and keep them ready for immediate use.
- (3) Every railway servant shall see that the railway servants in his charge concerned with the use of warning signals as prescribed by special instructions under rule 3.65, have a correct knowledge of their use.”.

H. DEFECTIVE FIXED SIGNALS AND POINTS

3.68 DUTIES OF STATION MASTER GENERALLY WHEN A SIGNAL IS DEFECTIVE.—

- (1) As soon as a Station Master becomes aware that any signal has become defective or has ceased to work properly, he shall—
 - (a) immediately arrange to place the signal at ‘on’ if it is not already in that position.
 - (b) depute competent railway servants with such hand signals and detonators as may be required to give signals at the foot of the defective signal, until he is satisfied that such signal has been put into proper working order:
 - (c) take action in accordance with Rules 3.69 and 3.70 as may be required for movement of trains past the defective signals; and
 - (d) report the occurrence to the railway servant responsible for the upkeep of the signals and if the section is controlled, the Controller also.
- (2) When the Station Master receives information of any defect in a signal not pertaining to his station from the Loco Pilot or the Guard or any other railway servant, he shall immediately inform the Station Master concerned of the fact and keep the controller advised, where the section is controlled.

- (3) In case of signals becoming defective at station situated on Centralised Traffic Control territories, the Centralised Traffic Control Operator on becoming aware of such defects, shall take action in accordance with special instructions.

3.69 DUTIES OF STATION MASTER WHEN AN APPROACH STOP SIGNAL IS DEFECTIVE.—

- (1) In the event of an Outer or a Home or a Routing signal becoming defective, the Station Master shall advise the station in rear and the nominated station in rear, same in a case where a signal post telephone or a Calling-on signal is provided on the defective signal, in order that the Loco Pilots of approaching train may be warned of the defective signal and issued a written authority to pass such signal on receipt of proceed hand signal at the foot of the defective signal.
- (2) The Station Master in rear as referred to in sub-rule (1), on receiving the advice of the defective signal, shall immediately acknowledge it and advise the Station Master of the Station where the signal has become defective, of the number of the first train which will be notified of the defective signal and again on receipt of the advice that the defective signal has been put into proper working order, shall advise the number of the train so notified last.
- (3) The Station Master of the station where the signal has become defective shall, before authorising a train to pass the defective signal, ensure that the conditions for taking 'off' that signal have been fulfilled. He shall then authorize the Loco Pilot to pass the defective signal at 'on' in one of the following manners: -
- (a) When the Loco Pilot of an approaching train has been advised of the defective signal at a station in rear—by deputing a competent railway servant in uniform under clause (b) of sub-rule (1) of Rule 3.68, to exhibit Proceed hand signal at the foot of the defective signal to the approaching train, In such cases, the Station Master shall not give Line Clear to the station in rear unless the conditions for taking 'off' the signal which has become defective, have been complied with; or
- (b) When the Loco Pilot of an approaching train has not been advised of the defective signal at station in rear- by issuing a written authority, authorising the Loco Pilot to pass the

defective signal at 'on', delivered at the foot of the defective signal through a competent railway servant; or

- (c) By taking 'off' the Calling-on signal where provided; or
 - (d) By authorising the Loco Pilot to pass the defective signal at 'on' over the signal post telephone where provided, in accordance with special instructions.
- (4) When the Home signal becomes defective, the Outer shall also be deemed to be out of order and the procedure prescribed in sub-rules (1), (2) and (3) shall be followed.

3.70 DUTIES OF STATION MASTER WHEN A DEPARTURE STOP SIGNAL IS DEFECTIVE.—

- (1) In the event of Starter becoming defective, the Station Master may authorize the Loco Pilot to pass such signal by a written authority which shall be handed over to the Loco Pilot at the station where the defective signal is located and in addition thereto, a competent railway servant shall show hand signals to the departing train in accordance with the instructions of the Station Master or by taking 'off' the Calling on signal, if provided under sub-rule (2) or Rule 3.13, after the train has been brought to a stand at the defective signal.
- (2) In the event of an Advanced Starter becoming defective, hand signals may be dispensed with and the Station Master may authorize the Loco Pilot to pass such signal by a written authority, which shall be handed over to the Loco Pilot at the station, where the defective signal is located:

Provided that in exceptional circumstances, where under approved special instructions, an advanced Starter protects any points hand signals shall not be dispensed with.

- (3) For the purpose of handing over the written authority mentioned in sub-rules (1) and (2), the train shall be stopped at the station where the defective signal is located. The written authority to pass a defective departure Stop signal shall not be handed over to the Loco Pilot, unless all the conditions for taking 'off' such signal have been fulfilled.

- (4) Where under approved special instructions a Calling-on signal has been provided below a departure Stop signal, other than the last Stop signal, the Calling-on signal shall not be taken 'off', unless the conditions for taking 'off' the departure Stop signal above it, have been fulfilled.

3.71 WARNER OR DISTANT SIGNALS DEFECTIVE IN THE 'OFF' POSITION.—

- (1)(a) If a Warner signal on a post by itself or a Distant signal is out of order and cannot be kept in the 'on' position, a Stop hand signal shall be shown at the foot of the signal. At night, the light or lights of the signal shall be extinguished and the train, after being first brought to a stand, may then be hand-signalled past the signal. Advice of the defective signal shall be given to the Loco Pilots of trains at the station in rear, warning them to stop at such signal.
- (b) If Warner signal placed below a Stop signal becomes defective and cannot be kept in the 'on' position, the Stop signal above it shall be treated as defective and by night the light of the Warner signal shall be extinguished.
- (2) If the Warner or Distant signal of an Intermediate Block Post is defective and cannot be kept in the 'on' position, the Intermediate Block Stop signal shall also be kept at 'on' and treated as defective and action taken as per Rule 3.75.

3.72 WARNER NOT TO BE USED WHEN STOP SIGNAL IS DEFECTIVE.—Whenever a Stop signal is defective or ceases to work properly at a station provided with Warners the Warner applying to the line to which the defective Stop signal applies shall be kept at 'on' until the defective Stop signal is rectified.

3.73 PASSING OF A GATE STOP SIGNAL AT 'ON'.—

- (1) When a Loco Pilot finds a gate Stop signal at 'on' he shall sound the prescribed code of whistle and bring his train to a stop in rear of the signal.
- (2)(a) If the gate Stop signal is provided with a 'G' marker, the Loco Pilot shall wait at the signal for one minute by day and two minutes by night and if the signal is not taken 'off' within this period, he may draw his train ahead cautiously up to the level crossing, and

- (b) if the Gateman is available and exhibiting hand signals, proceed further past the gate cautiously, or
- (c) if the Gateman is not available, or, is available but not exhibiting hand signals, he shall stop short of the level crossing, where he shall then be hand-signalled past the gate by the Gateman, if there is one or in the absence of a Gateman, by one of the members of the engine crew of the train after ascertaining that gates are closed against road traffic.
- (3) If the Loco Pilot finds, after stopping at the signal, that there is no 'G' Marker, he shall proceed further only in accordance with the procedure laid down under special instructions.

3.74 ABSENCE OF A FIXED SIGNAL OR A SIGNAL WITHOUT A LIGHT.—

- (1)(a) If there is no fixed signal at a place, where a fixed signal is ordinarily shown, or
- (b) If the light of a signal is not burning, when it should, or
- (c) If a white light is shown in place of a colour light, or
- (d) If the aspect of a signal is misleading or imperfectly shown, or
- (e) If more than one aspect is displayed,

The Loco Pilot shall act as if the signal was showing its most restrictive aspect:

Provided that during night, if in the case of a semaphore Stop Signal for approaching trains only, the Loco Pilot finds the signal light extinguished, he shall bring his train to a stop at such signal. If he finds that the day aspect of such signal is clearly visible and is satisfied that the signal is in the 'Off' position, he shall proceed past it up to the station cautiously at restricted speed obeying all intermediate Stop signals, if any, relating to him and report the matter to the Station Master for necessary action.

- (2) At station equipped with a colour light signal provided with a 'P' marker, the Loco Pilot shall bring his train to a stand if it

does not show any light or shows an imperfect aspect and having satisfied himself that the signal is provided with a 'P' marker, shall proceed preparing to stop at the next Stop signal and shall be guided further by its aspect.

3.75 PASSING OF INTERMEDIATE BLOCK STOP SIGNAL AT 'ON'. —

- (1) When a Loco Pilot finds an intermediate Block Stop signal at 'on' he shall stop his train in rear of the signal and contact the Station Master of the block station in rear on the telephone, if provided on the signal post.**
- (2) The Station Master shall authorise the Loco Pilot to pass the Intermediate Block Stop signal, if defective as prescribed by special instructions.**
- (3) If the telephone is not provided or is out of order, the Loco Pilot after waiting for 5 minutes at the signal shall pass it at 'on' and proceed cautiously and be prepared to stop short of any obstruction, at a speed not exceeding 15 Kilometres an hour if he has a good view of the line ahead, otherwise at a speed not exceeding 8 kilometres an hour and report the failure to the Station Master at the block station ahead.**
- (4) The Station Master of the block station working the Intermediate Block Stop signal on becoming aware that such a signal is defective shall, before despatching a train, treat the entire section up to the block station, immediately ahead of the Intermediate Block Post as one block section and issue a written authority to the Loco Pilot to pass the defective Intermediate Block Stop signal at 'on' without stopping at the signal, in accordance with the procedure prescribed by special instructions.**

3.76 INTIMATION TO OFFICIALS WHEN DEFECTS REMEDIED.— As soon as a defective signal has been put into good working order, the Station Master shall intimate the fact to the officials who were advised of its being defective.

3.77 DEFECTIVE OR DAMAGED POINTS, ETC.—

- (1) Whenever points, crossings or guard rails are defective or damaged, the railway servant in charge of operation of points**

shall protect them and immediately arrange to report the circumstances to the Station Master.

- (2) The Station Master, on becoming aware of such defective or damaged points, etc., shall.—
 - (a) immediately arrange to have the defect rectified by the railway servant responsible for their maintenance.
 - (b) arrange to ensure the safe passage of trains and
 - (c) keep the signal or signals concerned at 'on' until the defect is rectified.

3.78 DUTIES OF ENGINE CREW IN RESPECT OF SIGNALS.—

- (1)(a) The Loco Pilot shall pay immediate attention to and obey every signal whether the cause of the signal being shown is known to him or not;
 - (b) the Loco Pilot shall not, however, trust entirely to signal, but always be vigilant and cautious.
- (2)(a) The Loco Pilot shall whistle intermittently when his engine explodes detonator(s) and take every possible caution including reduction of speed as necessary, so as to have the train well under his control and be able to stop short of any obstruction on the line;
 - (b) after proceeding 1.5 kilometres from the place where his engine exploded detonator(s), if his engine does not explode any more detonator(s), he may then resume authorized speed; and
 - (c) report the incident to the next station or cabin.
- (3) If in consequence of fog, storm or any other reason, the view of the signal is obstructed, the Loco Pilot shall take every possible precaution, so as to have the train well under control.
- (4) When the Loco Pilot notices a signal warning of an obstruction, except detonator(s), he shall stop his train immediately and act on advice of the person exhibiting warning signal or on the basis of obstruction noticed by him.

- (5) In case no further details of exhibition of warning signal are noticed, after stopping for one minute by day and two minutes by night to ascertain the location and/or cause of the warning, he shall proceed cautiously upto the next block station, keeping a sharp look out.
- (6) The Loco Pilot shall acquaint himself with the system of working, location of signals and other local conditions affecting the running of trains on a section or sections of the railway over which he is to work and if he is not so acquainted with any portion of the railway over which he is to work, obtain the services of a qualified railway servant who is conversant with it to assist him.

3.79 DUTIES OF LOCO PILOT IN RESPECT OF A CALLING-ON SIGNAL.—The Loco Pilot of a train shall be guided always by the indication of the Stop signal below which the Calling-on signal is fixed. If this Stop signal is at 'on', he shall bring his train to a stop. If he finds that the Calling-on signal is taken 'off', he shall, after bringing his train to a stop, draw ahead with caution and be prepared to stop short of any obstruction.

3.80 DUTIES OF LOCO PILOT WHEN AN APPROACH STOP SIGNAL IS 'ON' OR DEFECTIVE.—

- (1) The Loco Pilot of a train shall not pass an Outer, a Home or a Routing signal that refers to him, when it is 'on' or defective, unless.—
 - (a) he has, at a previous station, received notice in writing specifying that the signal is out of order and unless he also receives a Proceed hand signal from a railway servant in uniform at the foot of such signal; or
 - (b) after coming to a stand, he is either given a written authority by the Station Master to proceed past such signal or is authorised by a Calling-on signal in the 'off' position or is authorised by the Station Master over the signal post telephone in accordance with special instructions.
- (2) The Loco Pilot of a train while passing an Outer, a Home or a Routing signal, when it is 'on' or defective, shall ensure that the speed of his train does not exceed 15 kilometres an hour.

3.81 DUTIES OF LOCO PILOT WHEN A DEPARTURE STOP SIGNAL IS ‘ON’ OR DEFECTIVE.—

- (1) The Loco Pilot of a train shall not pass a departure Stop signal that refers to him, when it is ‘on’ or defective, unless his train has been brought to a stop at the station, where the defective signal is situated and he is authorised to do so—**

 - (a) By a written permission from the Station Master, in addition, in the case of a starter, or Advanced starter protecting points, he shall not pass such signals, when ‘on’ or defective, unless he also receives a ‘proceed’ hand signal from a duly authorised member of the station staff, posted at the signal, or**
 - (b) By taking ‘off’ the calling-on signal, if provided under approved special instructions, vide sub-rule (2) of Rule 3.13.**
- (2) In the case of a last Stop signal, he shall not pass such signal, when ‘on’ or defective, unless he is also in possession of a proper authority to proceed under the system of working.**

3.82 PERMISSION BEFORE ENTERING ON OR CROSSING A RUNNING LINE.—No Loco Pilot shall take his engine on or across any running line until he has obtained the permission of the Station Master and has satisfied himself, that all the correct signals have been shown.

3.83 ASSISTANCE OF THE ENGINE CREW REGARDING SIGNALS.—

- (1) The Loco Pilot and the Assistant Loco Pilot shall identify each signal affecting the movement of the train as soon as it becomes visible. They shall call out the aspects of the signals to each other.**
- (2) The Assistant Loco Pilot shall, when not otherwise engaged, assist the Loco Pilot, in exchanging signals, as required.**
- (3) The provisions of sub-rules (1) and (2) shall, in no way, absolve the Loco Pilot of his responsibility in respect of observance of and compliance with the signals.**

3.84 DUTIES OF LOCO PILOTS AS TO SIGNAL WHEN TWO OR MORE ENGINES ARE ATTACHED TO TRAIN.—When

two or more engines are attached to a train, the Loco Pilot of the leading engine shall be responsible for the observance of and compliance with the signals and the Loco Pilot or Loco Pilots of other engine or engines shall watch for and take signals from the Loco Pilot of the leading engine, except in case, where special instructions are issued to the contrary.

3.85 REPORTING OF DEFECTS IN SIGNAL.—

- (1) Should a Loco Pilot or a Guard observe that a signal is rendered imperfectly visible by branches of trees or by any other cause, or that a signal light is partially obscured or not burning brightly enough to give a clear aspect, he shall report the matter to the Station Master at the next station, at which the train stops.**
- (2) When such a report is made by a Loco Pilot or a Guard, the Station Master shall take immediate steps to advise the Station Master concerned who shall get it rectified.**

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